Chemistry GCE A Level 2025 - Paper 1

Duration: 1h30 | Cameroon GCE Board

GENERAL CERTIFICATE OF EDUCATION BOARD, General Certificate of Education Examination. 0715 CHEMISTRY 1, ADVANCED LEVEL, JUNE 2025. Multiple choice question paper. Duration: one and a half hours (1h 30). Answer ALL the 50 questions in this examination. All questions carry equal marks. Non-programmable calculators are allowed. Each question has FOUR suggested answers: A, B, C and D. Mark only one answer for each question (if more than one answer is marked, the question scores zero). Avoid spending too much time on any one question.

Questions 1 - 35 (thirty five questions)

Directions: Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case.

Questions 36 - 45 (ten questions)

Directions: For each of the questions below, one or more responses is (are) correct. Decide which of the responses is (are) correct. Then choose:

A	B	C	D
1, 2, 3 correct	1, 2 only	2, 3 only	3 only

- **A** if 1, 2 and 3 are all correct
- **B** if 1 and 2 only are correct
- **C** if 2 and 3 only are correct
- **D** if 3 only is correct

Questions 46 - 50 (five questions)

Directions: Each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether the first statement is true or false. Decide whether the second statement is true or false. Then choose:

- **A** if both statements are true and the second statement is a CORRECT explanation of the first statement.
- **B** if both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
- **C** if the first statement is true, but the second statement is false.
- **D** if the first statement is false, but the second statement is true.

	First statement	Second statement	
A	True	True	Second statement is a CORRECT explanation of the first
B	True	True	Second statement is NOT a CORRECT explanation of the first

	First statement	Second statement	
C	True	False	
D	False	True	

1. Choose from the options a compound of the group 14 (IV) elements that will react readily with both aqueous NaOH and dilute HCl?

a) CO
b) $\mathrm{CO_2}$
c) $\mathrm{SiO_2}$
d) PbO

2. Which of the following substances is suitable for use as a primary standard?

a) $\mathrm{Ca(OH)_2}$
b) NaOH
c) $\mathrm{Na_2CO_3}$
d) $\mathrm{CaCO_3}$

3. Identify the compound which will form an alcohol on boiling with aqueous sodium hydroxide.

a) $\mathrm{CH_3COOH}$
b) $\mathrm{C_6H_5CH_2OH}$
c) $\mathrm{CH_3CH = CH_2}$
d) $\mathrm{CH_3CH_2Cl}$

4. Choose the correct expression for the partial pressure of methanol in a mixture comprising 1.75 moles of methanol and 1.25 moles of ethanol given that the vapour pressure of pure methanol is 92 mmHg and vapour pressure of pure ethanol is 67 mmHg.

a) $\frac{1.25}{3} \times 67$
b) $\frac{1.75}{3} \times 92$
c) $\frac{1.25}{1.75} \times 92$
d) $\frac{1.75}{1.25} \times 67$

5. Indicate the type of isomerism that exist between: $\mathrm{CH_3\text{-}O\text{-}CH_3}$ and $\mathrm{CH_3CH_2OH}$

a) Position isomerism
b) Optical isomerism
c) Geometric isomerism
d) Functional group isomerism

6. What will be the number of moles of bicarbonate ions ($\mathrm{HCO_3^-}$) in 24.6 g of $\mathrm{Ca(HCO_3)_2}$ (RMM $\mathrm{Ca(HCO_3)_2} = 122$)

a) 0.4
b) 0.2

c) 2.43×10^{23}
d) 1.21×10^{23}

7. Identify the particle X in the following nuclear reaction:

$$^{27}_{13}\mathrm{Al} + ^{1}_{0}\mathrm{n} \rightarrow \mathrm{X} + ^{4}_{2}\mathrm{He}$$

a) $^{24}_{10}\mathrm{Ne}$
b) $^{24}_{12}\mathrm{Mg}$
c) $^{24}_{11}\mathrm{Na}$
d) $^{23}_{11}\mathrm{Na}$

8. Which of the following pairs of compounds would undergo a condensation reaction?

a) water/ethanol
b) Phenol/Phenylamine
c) Ethanal/ethylamine
d) ethanol/2,4-DNPH

9. Calculate the volume of 0.5 M NaOH which is needed to completely neutralise $25.0\ \mathrm{cm}^3$ of 0.25 M HCl.

a) $6.25\ \mathrm{cm}^3$
b) $12.5\ \mathrm{cm}^3$
c) $25\ \mathrm{cm}^3$
d) $50\ \mathrm{cm}^3$

10. Transition metals form complexes with ligands because

a) They have lone pairs of electrons
b) They have catalytic properties
c) They have high charge density
d) They have low energy empty orbitals

11. What is the pH of a 0.2 M NaOH solution

a) 13.3
b) 12.4
c) 1.6
d) 0.7

12. Which spectroscopic technique can be used to identify Protons in an organic compound?

a) Infra-red spectroscopy
b) Ultra-violet spectroscopy
c) Nuclear magnetic resonance spectroscopy
d) Mass spectroscopy

13. The maximum oxidation number of an element with electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$ is:

a) +6
b) +5
c) +3
d) +2

14. An organic compound, A, of molecular formula $C_4H_{10}O$ undergoes oxidation to give a compound B, of molecular formula $C_4H_8O_2$. Compound A could be:

a) $CH_3CH(OH)CH_3$
b) $(CH_3)_3COH$
c) $(CH_3)_2CHCH_2OH$
d) $CH_3CH_2CH(OH)CH_3$

15. Given the half cells:

$Ag^+(aq) + e^- \rightleftharpoons Ag(s); E^\ominus = +0.80\ V$

and $Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s); E^\ominus - 0.76\ V$

Calculate the emf of the cell.

a) $+1.56\ V$
b) $-1.56\ V$
c) $+0.04\ V$
d) $-0.04\ V$

16. Consider the rate equation: $Rate = k[A]^2[B]$. Suggest the overall order of the reaction.

a) 0
b) 1
c) 2
d) 3

17. Which one of the following families of elements will dissolve in water to give a solution of low pH?

a) Transition elements
b) Halogens
c) Alkali metals
d) Group IV elements

18. Identify from the following, a reaction in which Sulphur is oxidized only.

a) $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$
b) $Fe(s) + S(s) \rightarrow FeS(s)$
c) $Na_2SO_3(aq) + S(s) \rightarrow Na_2S_2O_3(aq)$
d) $H_2(g) + S(s) \rightarrow H_2S(g)$

19. State the units for the equilibrium process: $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$.

a) $mol\ dm^{-3}$
b) $mol^2\ dm^{-6}$
c) $mol^{-2}\ dm^6$
d) $mol^{-1}\ dm^3$

20. What is the IUPAC name for the compound $CH_3CH(CH_3)CH(CH_2CH_3)CH_2CH_2CH_3$.

a) 3-ethyl-2-methyl hexane
b) 4-ethyl-5-methyl hexane

c) 3-isopropyl hexane
d) 2-methyl-3-propyl pentane

21. From the energy diagram below, calculate the value of X in kJ.

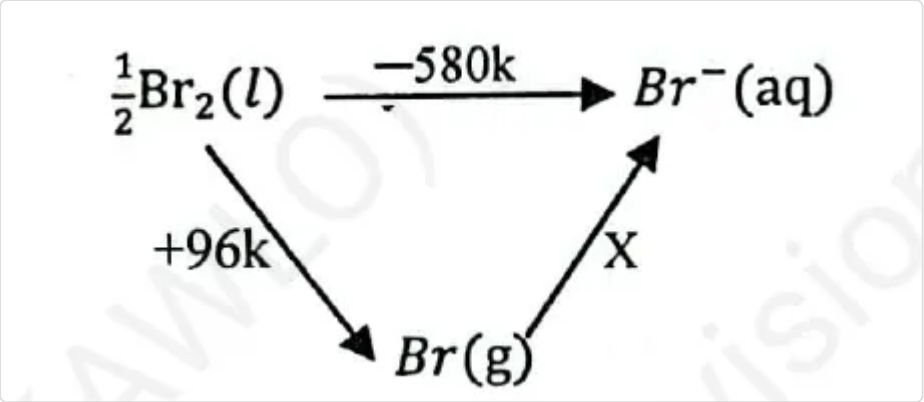


Figure as printed: a triangle with $\frac{1}{2}\mathrm{Br}_2(l)$ at the top left and $\mathrm{Br}^-(aq)$ at the top right, joined by an arrow labelled -580 k (kJ); an arrow labelled $+96$ k (kJ) from $\frac{1}{2}\mathrm{Br}_2(l)$ down to $\mathrm{Br}(g)$; and an arrow labelled X from $\mathrm{Br}(g)$ up to $\mathrm{Br}^-(aq)$.

a) -676
b) +676
c) -484
d) +484

22. Select the carboxylic acid which can form a silver mirror when heated with Tollen's reagent.

a) Ethanoic acid
b) Methanoic acid
c) Benzoic acid
d) Propanoic acid

23. Identify the molecule with the most polar covalent bond.

a) HBr
b) HI
c) HCl
d) HF

24. Which of the following reactions could be used to prepare gaseous hydrogen iodide in the laboratory?

a) pass $\mathrm{H_2(g)}$ over heated $\mathrm{I_2(s)}$
b) warm conc. $\mathrm{HNO_3}$ with $\mathrm{I_2(s)}$
c) warm conc. $\mathrm{H_3PO_4}$ with $\mathrm{NaI(s)}$
d) warm conc. $\mathrm{H_2S}$ with NaI and $\mathrm{MnO_2}$

25. The Hoffmann degradation reaction is characterised by:

a) Ring opening
b) Ring closure
c) Chain lengthening
d) Chain shortening

26. Which of these s-block nitrates decompose to give the nitrite and oxygen?

a) $LiNO_3$
b) KNO_3
c) $Ca(NO_3)_2$
d) $Mg(NO_3)_2$

27. A sample of an organic compound was analysed to contain 1.2 g of carbon, 0.3 g of hydrogen and 2.4 g of oxygen. Determine the empirical formula of the compound. (RAM: C = 12, H = 1, O = 16)

a) C_2H_3O
b) $C_2H_6O_2$
c) $C_2H_6O_3$
d) CH_3O

28. Identify the hydrohalic acid with the highest reducing power

a) HF
b) HCl
c) HBr
d) HI

29. Chloroethene ($CH_2=CHCl$) is a monomer used in producing a very important polymer commonly called:

a) Polychloroethane
b) Polyvinylchloride
c) Chloropolyethene
d) Polystyrene

30. What is the coordination number of cobalt in the complex: $[Co(EDTA)]^{2-}$

a) 1
b) 2
c) 4
d) 6

31. Determine the product (Y) of the reaction

$$C_6H_5NO_2 \xrightarrow[55^\circ C - 60^\circ C]{\text{Conc. } H_2SO_4/\text{Conc. } HNO_3} Y$$

a) 1,2-Dinitrobenzene
b) 1,3-Dinitrobenzene
c) 1,4-Dinitrobenzene
d) 1,2,3-Trinitrobenzene

32. Identify the structure of aminoethanoic acid (NH_2CH_2COOH) in acidic medium.

a) $^+NH_3CH_2COOH$
b) NH_2CH_2COOH
c) $^+NH_3CH_2COO^-$
d) $NH_2CH_2COO^-$

33. Select a compound of nitrogen which produces a mixture of acids when dissolved in water.

a) HNO_3
b) N_2O_5
c) NO_2
d) NH_3

34. Which of the following is an amphoteric oxide in period 3 of the periodic table?

a) BeO
b) MgO
c) Al_2O_3
d) SiO_2

35. The element X (where X is not the usual symbol of the element) has electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$. Identify the group and period to which X belongs.

a) group III, period 3
b) group III, period 5
c) group V, period 3
d) group V, period 2

36. Which of the following statement(s) is/are true about the nitrate ion, NO_3^- ?

1. it has a trigonal planar shape
2. its bond angle is 120°
3. it has a trigonal pyramidal shape

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

37. Select the statement(s) that is/are characteristic of ideal mixtures.

1. Obey Raoult's law
2. Have components with similar chemical structures
3. Have components with different intermolecular forces

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

38. Amides can be prepared by:

1. Dehydration of ammonium salts of carboxylic acids
2. Reaction of ammonia with acid chlorides
3. Reaction of ammonia with esters

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

39. The reactivity of group 1 elements increases down the group because:

1. Ionization energy gets lower
2. Effective nuclear charge reduces
3. The outer electrons get more loosely bound.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

40. Propanone (CH_3COCH_3) and propanal (CH_3CH_2CHO) differ in their reactions with:

1. phosphorus pentachloride
2. 2,4- dinitrophenylhydrazine
3. ammoniacal silver nitrate

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

41. Select the hydride(s) which when added to pure water will increase the pH

1. NH_3
2. NaH
3. HBr

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

42. Iodine ions are oxidised by peroxydisulphate ions according to the equation $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$.

Which method can be used to monitor rate?

1. Titrimetry
2. Colorimetry
3. Dilatometry

Options as printed: A) 1, 2 and 3 are all correct B) 1 and 2 only are correct C) 2 and 3 only are correct D) 3 only is correct

43. When phenylethanoate ($CH_3COOC_6H_5$) is hydrolysed by excess $NaOH(aq)$, the products of the reaction are:

1. phenoxide ion and ethanol
2. benzoate ion and ethanol
3. phenol and ethanoate ion

Options as printed: A) 1, 2 and 3 are all correct B) 1 and 2 only are correct C) 2 and 3 only are correct D) 3 only is correct

44. Choose the statement(s) that is/are true with respect to HBr.

1. It is prepared by the reaction of concentrated phosphoric acid with an alkali metal bromide.

2. Its boiling point is higher than that of HCl.
3. It is a weaker acid than HI

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

45. In an exothermic reaction,

1. Heat is released to the surroundings
2. Energy of reactants is higher than energy of products
3. Heat is absorbed from the surroundings.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 3 only is correct

First statement: Catenation is extensive in carbon but limited in silicon.

Second statement: Carbon forms stronger C-C bonds than silicon due to its small size.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Nitrogen gas is evolved when R_2NH reacts with nitrous acid at $< 5^{\circ}\mathrm{C}$

Second statement: The reaction of $\mathrm{RNH_2}$ with nitrous acid can be used to distinguish it from $\mathrm{R_2NH}$.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Atomic radius decreases across period 2 of the periodic table

Second statement: Across period 2 of the periodic table, effective nuclear charge decreases as electrons are added to the same shell.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: 2-hydroxybenzoic acid has a lower boiling point than 4-hydroxybenzoic acid

Second statement: The hydroxyl and carboxyl groups in 2-hydroxy benzoic acid are close to each other, hence form intramolecular hydrogen bonds while in 4-hydroxybenzoic acid, they are far apart and form intermolecular hydrogen bonds.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

First statement: Scandium is not considered a transition metal.

Second statement: Scandium has a completely filled 3d sub-shell.

a) Both statements are true and the second statement is a CORRECT explanation of the first statement.
b) Both statements are true and the second statement is NOT a CORRECT explanation of the first statement.
c) The first statement is true, but the second statement is false.
d) The first statement is false, but the second statement is true.

Want to check your answers?

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